MANSI GANGWAR

Senior Software Engineer
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Senior Software Engineer with over 8 years of experience designing and scaling distributed backend systems using Java, Spring Boot, and microservices architecture. Currently at SAP, leading development of multi-tenant enterprise systems and Kafka-based event-driven workflows processing high-volume data in distributed systems. Strong expertise in system design, API development, and performance optimization, reducing latency from around 1 second to under 200 milliseconds. Experienced in building resilient systems using caching, circuit breakers, and retry mechanisms, and leading teams to deliver high-impact features in cloud environments.

TECHNICAL SKILLS

Languages: Java, Python, C, C++, JavaScript

Frontend Technologies: React, JavaScript, HTML, CSS

Frameworks: Spring Boot, Spring MVC, Spring Data JPA, Hibernate

Architecture: Microservices, REST APIs, Distributed Systems, Event-Driven Architecture

Messaging: Apache Kafka, Zookeeper

Caching: Redis

Databases: SAP HANA, MySQL, Oracle, MongoDB

Cloud & DevOps: SAP BTP (Cloud Foundry), AWS (EC2, S3), Kubernetes (container orchestration), Docker, Jenkins, CI/CD

Tools: Git, Maven, Splunk, ServiceNow

Testing: JUnit, Mockito

Security: RBAC, OAuth2, JWT

Core Concepts: Concurrency, Multithreading, System Design, Performance Optimization, Connection Pool Design, Agile, Scrum, AI-assisted Development, Prompt Engineering

PROFESSIONAL EXPERIENCE

Senior Software Developer | SAP Labs — **Bangalore, India | December 2019 – Present**

- **Designed and led** development of the Dependents module in **SAP SuccessFactors**, used by **5 plus Fortune 500 clients**, handling large-scale employee data workflows in a multi-tenant SaaS environment.
- Architected and refactored backend systems into Dockerized **Spring Boot microservices**, improving modularity, scalability, and deployment efficiency.
- Designed event-driven microservices architecture using **Kafka** to decouple services and enable scalable asynchronous processing of high-volume background workflows, **handling millions of events per day** in multi-tenant systems.
- Designed backend services for the **Organizational Reorg module**, handling complex hierarchical transformations and orchestrating long-running workflows across distributed systems.
- Improved system performance and reliability by implementing caching, circuit breakers, and retry mechanisms, along with optimizing CI/CD pipelines, reducing deployment time by **30 percent** and latency from around **1 second to under 200 milliseconds**.
- Improved production reliability by resolving over **50 issues**, reducing high-severity incidents by 25 percent, and building observability dashboards using Splunk.
- Reviewed over **500 pull requests**, improving code quality, enforcing modular architecture, and standardizing API design practices across teams.
- Leveraged **AI-assisted tools (GitHub Copilot, Claude)** to accelerate development, debugging, and test generation, improving engineering productivity.
- **Led a team of 4 engineers**, owning architectural decisions, driving sprint execution, and delivering high-impact features in distributed systems.
- **Mentored over 10 engineers**, improving onboarding efficiency and overall team productivity.
- Consistently rated as a top performer, leading to rapid successive promotions based on technical and leadership contributions.

Associate Developer | Huawei **Bangalore, India| October 2017 - November 2019**

- Developed full-stack features using Spring Boot **REST APIs with jQuery/AJAX frontend, supporting agile releases (2/month).**
- Migrated legacy applications to Spring Boot, improving **maintainability, scalability**, and reducing **system complexity** across enterprise modules.
- Led **refactoring and backend optimization**, improving **modularity, reducing coupling**, and enhancing **API performance**.
- Partnered with product teams to define **data flows, service boundaries and reusable backend components** for distributed systems
- Designed and maintained **data access layers (MySQL, Oracle)** ensuring efficient and reliable data operations.
- Built **reporting dashboards (daily/weekly/ad-hoc)** supporting analytics, business operations, and customer insights.
- Ensured seamless **UI-backend integration** by defining and validating **API contracts**.
- **Best Capability Improvement Team Award (March 2019)**

EDUCATION

M.Tech (Hons.) Degree in Information Security **May 2015 - Jun 2017**
Birla Institute of Technology, Ranchi

B.Tech (Hons.) Degree in Computer Science **May 2010 - Jun 2014**
SRMS, Bareilly

INTERNSHIP

SAG LAB ,DRDO **New Delhi, India|July 2016 - July 2017**

Project: ***Analysis of Genetic Algorithm on Stream Cipher***

- Worked on encryption using LFSR Ciphers and decryption using Genetic Algorithm.
- Analyzed and compared results across various polynomial degrees to evaluate cipher strength and decryption accuracy.

RESEARCH

Published IEEE research on Genetic Algorithms, demonstrating strong analytical and problem-solving skills.

CERTIFICATION

Microsoft Certified: Azure AI Fundamentals (AI-900) **Sept 2023**
Validates AI and machine learning fundamentals, including Generative AI, computer vision, natural language processing, and responsible AI on Azure.